

Assignment 4 – DAX & Data Visualization

Calculated Columns:

- **Create a Calculated Column for 'Category Type':**

Go to 'New column' option in home tab to create new column. In the formula bar of the column use concatenation. Here, The formula used 'Category type = 'Order Details'[Category] & " " & 'Order Details'[Sub-Category]'

The screenshot shows the Power BI Desktop interface. The 'Column tools' ribbon is active, and the formula bar contains the DAX formula: `Category type = 'Order Details'[Category] & " " & 'Order Details'[Sub-Category]`. The data table 'Order Details' is displayed with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and the newly created 'Category type'. The 'Category type' column contains concatenated values of 'Category' and 'Sub-Category' separated by a space. The table caption indicates 1,500 rows and 17 distinct values for the 'Category type' column.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type
B-25602	561.00	212.00	3	Clothing	Saree	Clothing Saree
B-25602	119.00	-5.00	8	Clothing	Saree	Clothing Saree
B-25603	193.00	-166.00	3	Clothing	Saree	Clothing Saree
B-25604	157.00	5.00	9	Clothing	Saree	Clothing Saree
B-25605	75.00	0.00	7	Clothing	Saree	Clothing Saree
B-25609	25.00	-5.00	4	Clothing	Saree	Clothing Saree
B-25610	43.00	0.00	3	Clothing	Saree	Clothing Saree
B-25611	160.00	-59.00	2	Clothing	Saree	Clothing Saree
B-25613	1603.00	0.00	9	Clothing	Saree	Clothing Saree
B-25619	353.00	90.00	8	Clothing	Saree	Clothing Saree
B-25622	534.00	0.00	3	Clothing	Saree	Clothing Saree
B-25623	149.00	-87.00	4	Clothing	Saree	Clothing Saree
B-25625	635.00	-349.00	5	Clothing	Saree	Clothing Saree
B-25628	24.00	-9.00	4	Clothing	Saree	Clothing Saree
B-25633	711.00	-8.00	4	Clothing	Saree	Clothing Saree
B-25635	382.00	30.00	3	Clothing	Saree	Clothing Saree
B-25636	637.00	113.00	5	Clothing	Saree	Clothing Saree
B-25640	122.00	-47.00	4	Clothing	Saree	Clothing Saree
B-25646	20.00	-8.00	2	Clothing	Saree	Clothing Saree
B-25647	42.00	-6.00	4	Clothing	Saree	Clothing Saree
B-25648	55.00	-26.00	4	Clothing	Saree	Clothing Saree
B-25648	130.00	-41.00	4	Clothing	Saree	Clothing Saree
B-25650	211.00	-105.00	2	Clothing	Saree	Clothing Saree

- **Calculate Revenue per Order in 'Order Details' Table:**

Go to 'New column' option in home tab to create new column. In the formula bar of the column use the formula 'Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]'

The screenshot shows the Power BI Desktop interface with the 'Home' ribbon active. The formula bar contains the DAX formula: `Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]`. The data table 'Order Details' is displayed with an additional column, 'Revenue', which contains the calculated values for each row. The table caption indicates 1,500 rows and 845 distinct values for the 'Revenue' column.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type	Revenue
B-25602	561.00	212.00	3	Clothing	Saree	Clothing Saree	1683
B-25602	119.00	-5.00	8	Clothing	Saree	Clothing Saree	952
B-25603	193.00	-166.00	3	Clothing	Saree	Clothing Saree	579
B-25604	157.00	5.00	9	Clothing	Saree	Clothing Saree	1413
B-25605	75.00	0.00	7	Clothing	Saree	Clothing Saree	525
B-25609	25.00	-5.00	4	Clothing	Saree	Clothing Saree	100
B-25610	43.00	0.00	3	Clothing	Saree	Clothing Saree	129
B-25611	160.00	-59.00	2	Clothing	Saree	Clothing Saree	320
B-25613	1603.00	0.00	9	Clothing	Saree	Clothing Saree	14427
B-25619	353.00	90.00	8	Clothing	Saree	Clothing Saree	2824
B-25622	534.00	0.00	3	Clothing	Saree	Clothing Saree	1602
B-25623	149.00	-87.00	4	Clothing	Saree	Clothing Saree	596
B-25625	635.00	-349.00	5	Clothing	Saree	Clothing Saree	3175
B-25628	24.00	-9.00	4	Clothing	Saree	Clothing Saree	96
B-25633	711.00	-8.00	4	Clothing	Saree	Clothing Saree	2844
B-25635	382.00	30.00	3	Clothing	Saree	Clothing Saree	1146
B-25636	637.00	113.00	5	Clothing	Saree	Clothing Saree	3185
B-25640	122.00	-47.00	4	Clothing	Saree	Clothing Saree	488
B-25646	20.00	-8.00	2	Clothing	Saree	Clothing Saree	40
B-25647	42.00	-6.00	4	Clothing	Saree	Clothing Saree	168
B-25648	55.00	-26.00	4	Clothing	Saree	Clothing Saree	220
B-25648	130.00	-41.00	4	Clothing	Saree	Clothing Saree	520
B-25650	211.00	-105.00	2	Clothing	Saree	Clothing Saree	422

- **Create a Calculated Column to Categorize Sales :**

Go to 'New column' option in home tab to create new column. In the formula bar of the column use the formula 'Sales Category = if('Order Details'[Amount]>=AVERAGE('Order Details'[Amount]),"Above Average","Below Average")

The screenshot shows the Power BI Desktop interface. The 'Column tools' ribbon is active, displaying options for Name, Data type, Format, Summarization, Data category, Sort by, Data groups, Manage relationships, and New column. The formula bar shows the DAX formula for the 'Sales Category' column: `1 Sales Category = if('Order Details'[Amount]>=AVERAGE('Order Details'[Amount]),"Above Average","Below Average")`. The data table below shows the results of this calculation for 20 rows of order details.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type	Revenue	Sales Category
B-25602	561	212.00	3	Clothing	Saree	Clothing Saree	1683	Above Average
B-25602	119	-5.00	8	Clothing	Saree	Clothing Saree	952	Below Average
B-25603	193	-166.00	3	Clothing	Saree	Clothing Saree	579	Below Average
B-25604	157	5.00	9	Clothing	Saree	Clothing Saree	1413	Below Average
B-25605	75	0.00	7	Clothing	Saree	Clothing Saree	525	Below Average
B-25609	25	-5.00	4	Clothing	Saree	Clothing Saree	100	Below Average
B-25610	43	0.00	3	Clothing	Saree	Clothing Saree	129	Below Average
B-25611	160	-59.00	2	Clothing	Saree	Clothing Saree	320	Below Average
B-25613	1603	0.00	9	Clothing	Saree	Clothing Saree	14427	Above Average
B-25619	353	90.00	8	Clothing	Saree	Clothing Saree	2824	Above Average
B-25622	534	0.00	3	Clothing	Saree	Clothing Saree	1602	Above Average
B-25623	149	-87.00	4	Clothing	Saree	Clothing Saree	596	Below Average
B-25625	635	-349.00	5	Clothing	Saree	Clothing Saree	3175	Above Average
B-25628	24	-9.00	4	Clothing	Saree	Clothing Saree	96	Below Average
B-25633	711	-8.00	4	Clothing	Saree	Clothing Saree	2844	Above Average
B-25635	382	30.00	3	Clothing	Saree	Clothing Saree	1146	Above Average
B-25636	637	113.00	5	Clothing	Saree	Clothing Saree	3185	Above Average
B-25640	122	-47.00	4	Clothing	Saree	Clothing Saree	488	Below Average
B-25646	20	-8.00	2	Clothing	Saree	Clothing Saree	40	Below Average
B-25647	42	-6.00	4	Clothing	Saree	Clothing Saree	168	Below Average
B-25648	55	-26.00	4	Clothing	Saree	Clothing Saree	220	Below Average
B-25648	130	-41.00	4	Clothing	Saree	Clothing Saree	520	Below Average
B-25650	217	-105.00	2	Clothing	Saree	Clothing Saree	422	Below Average

Calculated Measures:

- **Calculate Order Count:**

Go to 'New measure' option in home tab to create new measure. In the formula bar of the measure use the formula 'Total Number of orders = COUNT('Order Details'[Order ID])'

The screenshot shows the Power BI Desktop interface. The 'Measure tools' ribbon is active, displaying options for Name, Home table, Format, Data category, and New/Quick measure measure. The formula bar shows the DAX formula for the 'Total Number of orders' measure: `1 Total Number of orders = COUNT('Order Details'[Order ID])`. The data table below shows the results of this calculation for 20 rows of order details.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type	Revenue	Sales Category
B-25602	561	212.00	3	Clothing	Saree	Clothing Saree	1683	Above Average
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B-25604	157	5.00	9	Clothing	Saree	Clothing Saree	1413	Below Average
B-25605	75	0.00	7	Clothing	Saree	Clothing Saree	525	Below Average
B-25609	25	-5.00	4	Clothing	Saree	Clothing Saree	100	Below Average
B-25610	43	0.00	3	Clothing	Saree	Clothing Saree	129	Below Average
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B-25650	217	-105.00	2	Clothing	Saree	Clothing Saree	422	Below Average

- **Calculate Average Profit in Delhi :**

Go to 'New measure' option in home tab to create new measure. In the formula bar of the measure use the formula 'Average Profit in delhi = **CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[State]="Delhi")**'

Table: List of Orders (560 rows) Columns: Average Profit in delhi (0 distinct values)

Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
B-25620	20-04-2018	Sarita	Maharashtra	Pune
B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
B-25623	22-04-2018	Atharv	West Benaol	Kolkata

- **Calculate Year-to-Date (YTD) Sales :**

Create a calendar table consisting of dates, import it to powerbi, Go to 'New measure' option in home tab to create new measure. In the formula bar of the measure use the formula 'sales YTD = **TOTALYTD(sum('Order Details'[Amount]),'Calendar'[Date])**

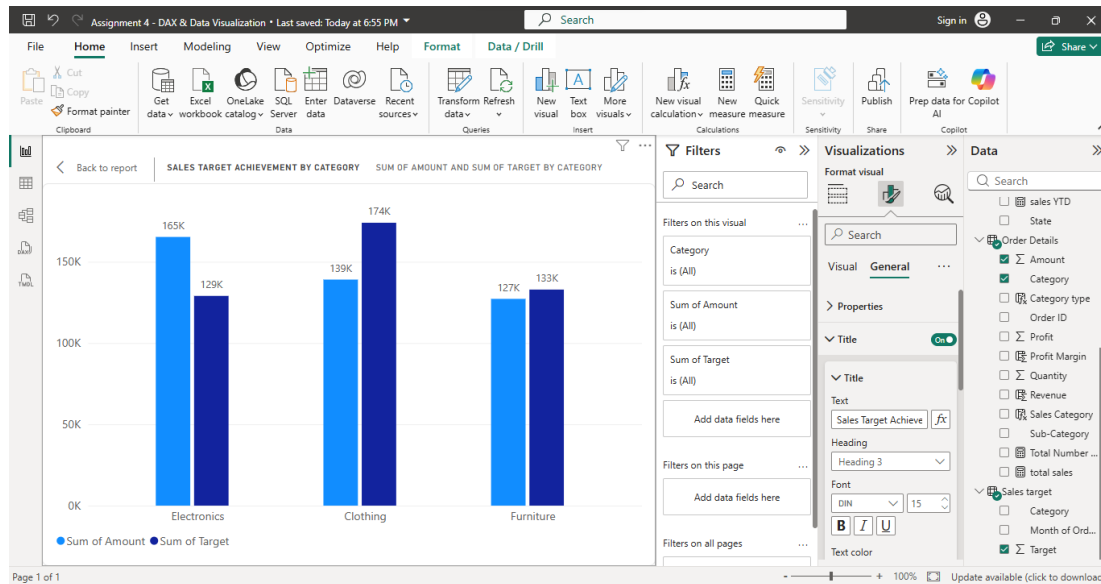
Table: List of Orders (560 rows) Columns: sales YTD (0 distinct values)

Order ID	Order Date	CustomerName	State	City
B-25601	Sunday, April 1, 2018	Bharat	Gujarat	Ahmedabad
B-25602	Sunday, April 1, 2018	Pearl	Maharashtra	Pune
B-25603	Tuesday, April 3, 2018	Jahan	Madhya Pradesh	Bhopal
B-25604	Tuesday, April 3, 2018	Divsha	Rajasthan	Jaipur
B-25605	Thursday, April 5, 2018	Kasheen	West Bengal	Kolkata
B-25606	Friday, April 6, 2018	Hazel	Karnataka	Bangalore
B-25607	Friday, April 6, 2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	Sunday, April 8, 2018	Aarushi	Tamil Nadu	Chennai
B-25609	Monday, April 9, 2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	Monday, April 9, 2018	Yogesh	Bihar	Patna
B-25611	Wednesday, April 11, 2018	Anita	Kerala	Thiruvananthapuram
B-25612	Thursday, April 12, 2018	Shrichand	Punjab	Chandigarh
B-25613	Thursday, April 12, 2018	Mukesh	Haryana	Chandigarh
B-25614	Friday, April 13, 2018	Vandana	Himachal Pradesh	Simla
B-25615	Sunday, April 15, 2018	Bhavna	Sikkim	Gangtok
B-25616	Sunday, April 15, 2018	Kanak	Goa	Goa
B-25617	Tuesday, April 17, 2018	Sagar	Nagaland	Kohima
B-25618	Wednesday, April 18, 2018	Manju	Andhra Pradesh	Hyderabad
B-25619	Wednesday, April 18, 2018	Ramesh	Gujarat	Ahmedabad
B-25620	Friday, April 20, 2018	Sarita	Maharashtra	Pune
B-25621	Friday, April 20, 2018	Deepak	Madhya Pradesh	Bhopal
B-25622	Sunday, April 22, 2018	Monisha	Rajasthan	Jaipur
B-25623	Sunday, April 22, 2018	Atharv	West Benaol	Kolkata

Data Visualization:

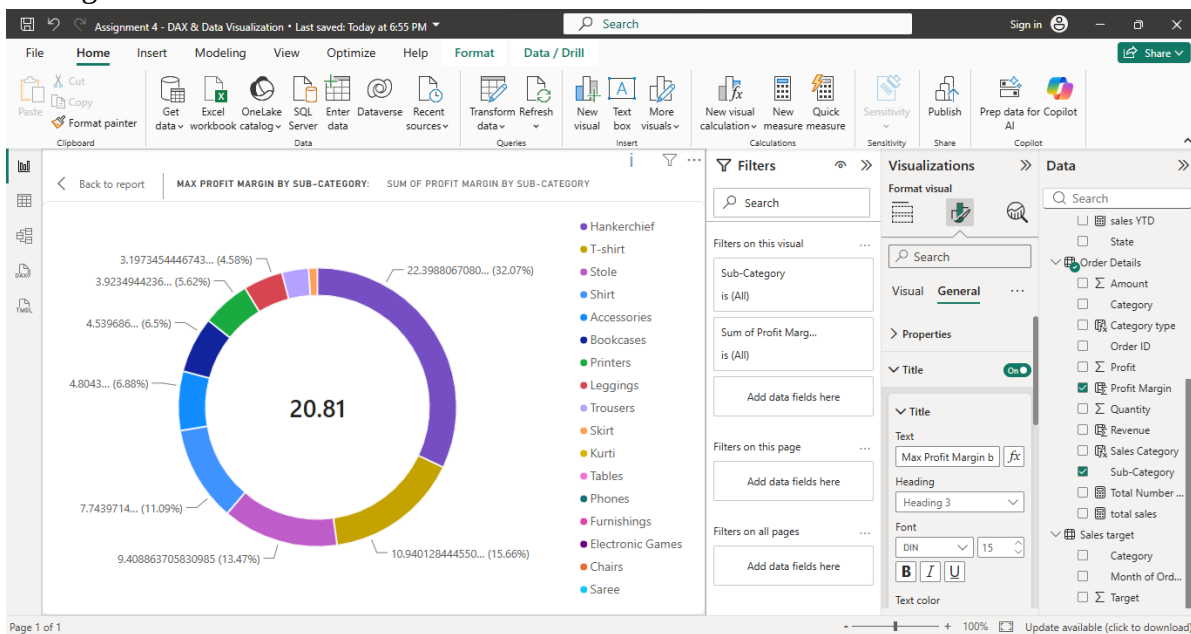
- **Sales Target Achievement by Category:**

In Report view, insert 'clustered column chart', on x – axis put 'category' and on y- axis put 'sum of amount' and 'sum of target'.



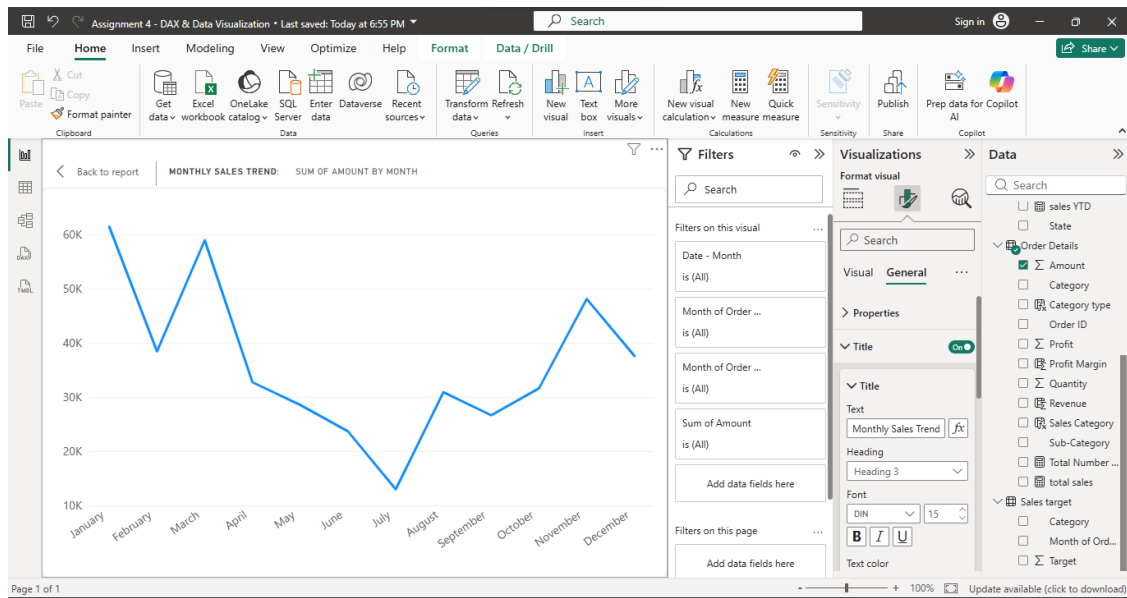
- **Max Profit Margin by Sub-Category:**

In report view, insert 'Donut Chart', On legend put 'Sub category' and on value put 'profit margin'.



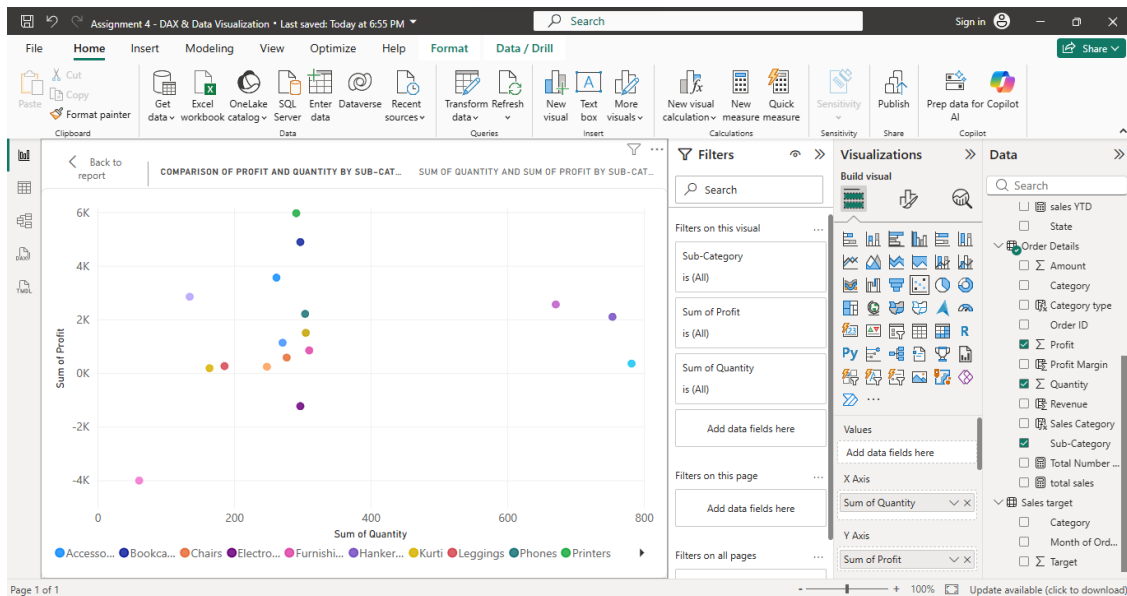
- **Monthly Sales Trend:**

In Report view, insert 'Line chart', on x axis put 'order date' expanded and remove day, year and quarter to get only the month and on y axis put amount.



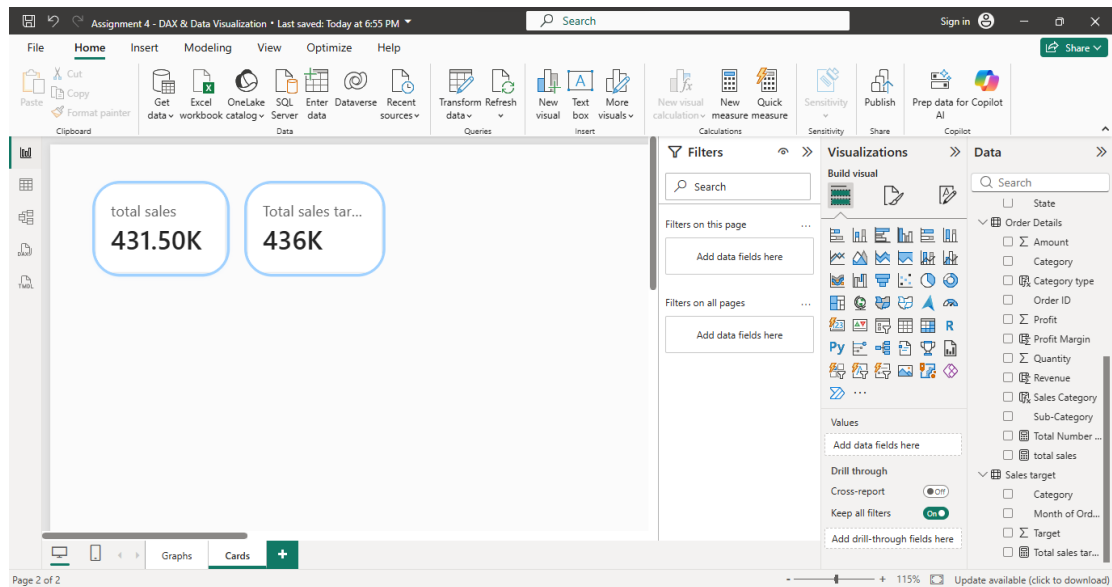
- **Comparison of Profit and Quantity by Sub-Category:**

In report view, insert 'scatter chart', on x - axis put 'quantity', on y-axis put 'profit' and put 'sub category' in legend to get the required scatterchart.

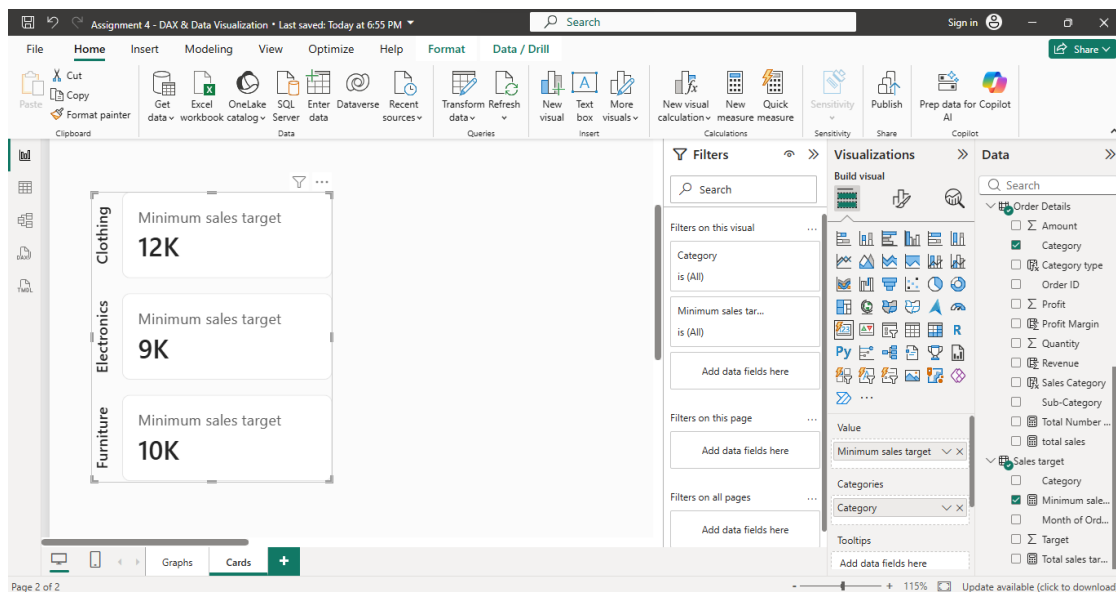


- **Comparison of Total Sales Amount and Target :**

In report view, insert 2 'cards' and add the 'total sales' and 'total sales target' measure on values of each card.

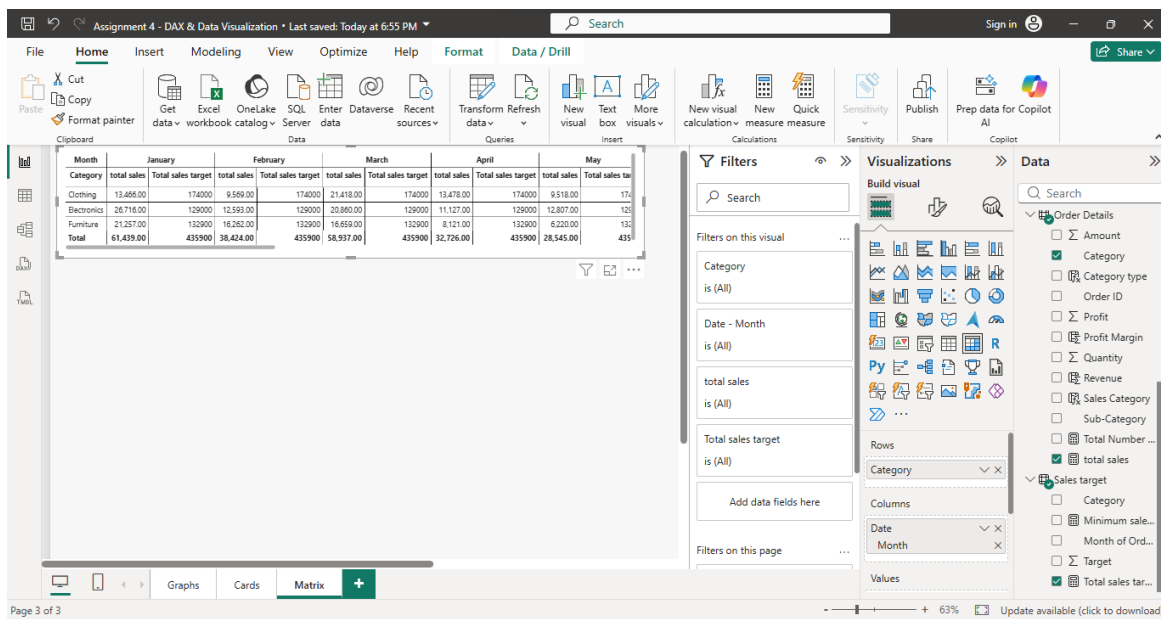


For multirow card to display the minimum target for each segment(category), insert 'cards' on category add 'category' and in values add measure of 'minimum sales target'.



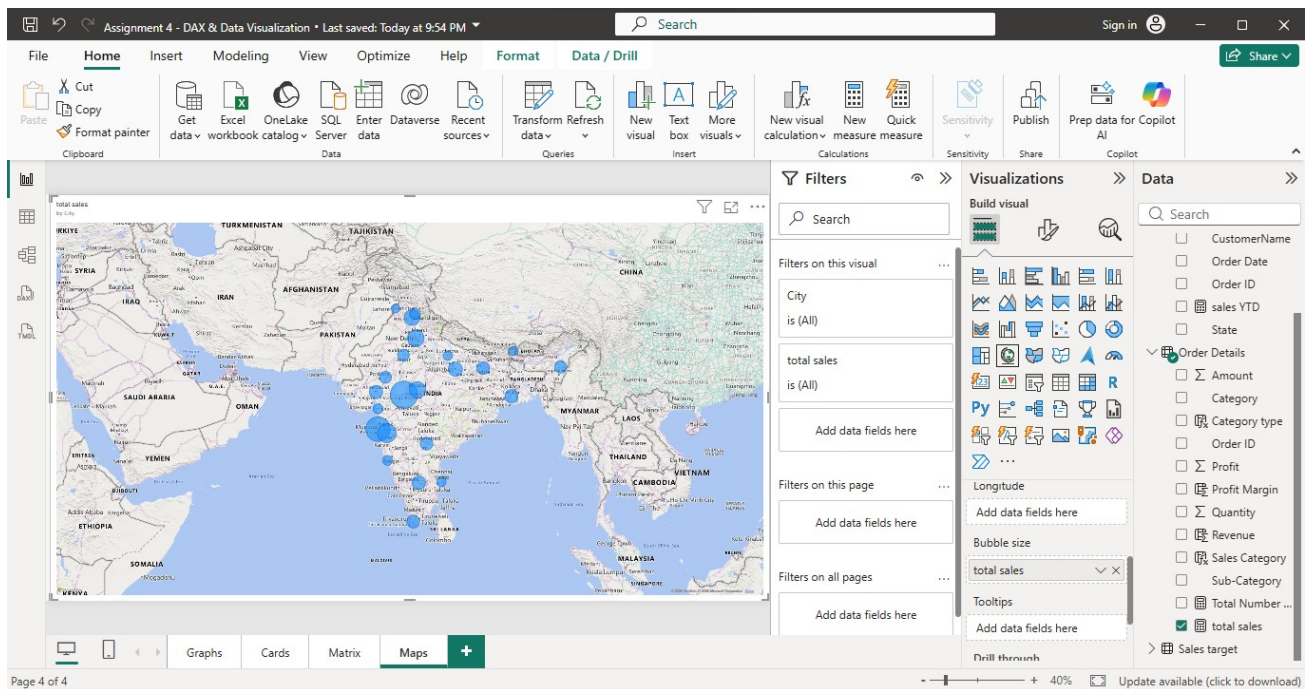
- **Sales Performance Matrix:**

In report view, insert matrix, put 'category' on 'rows', 'months' on 'columns', measure of 'total sales' and 'total sales target' on 'values'



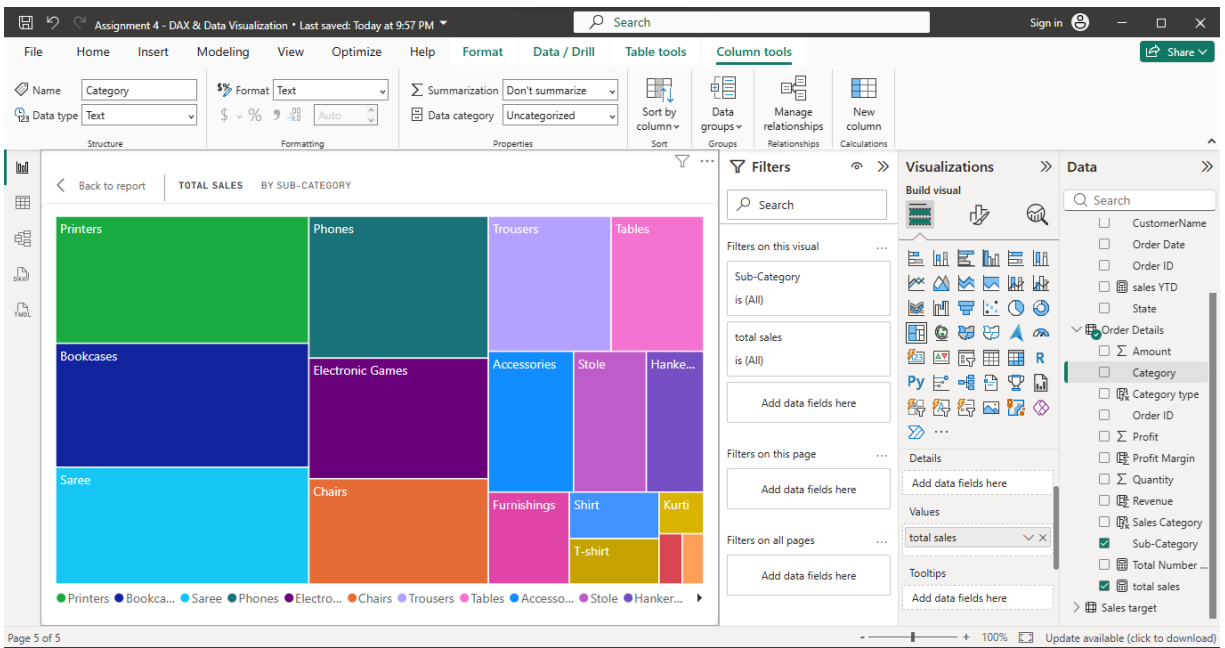
- **Geographic Sales Analysis:**

In report view, insert map, put 'city' on location and 'total sales' measure on bubble size.



- **Sales Distribution by Sub-Category:**

In report view, insert 'treemap', put 'sub category' on category and 'total sales' measure on value.



- **Order Count Analysis by State:**

In report view, insert 'funnel', on category put 'state' and on values put 'Total number of orders measure'.

